Umair Ul Hassan

Full-Stack Developer — Web & Mobile

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Full-Stack Developer building web and mobile applications across SaaS, commerce, ERP, and applied AI. Experienced with TypeScript/JavaScript, React, Next.js, Node.js, React Native, PostgreSQL, MongoDB, and Python-based AI services, with hands-on experience in API development, production UI, computer vision, AR, authentication, payments, and real-time systems.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Java
Web / Backend	React, Next.js, Node.js, Express.js, FastAPI, REST APIs
Mobile / 3D	React Native, Expo, Three.js, Viro React
AI / ML	PyTorch, Vision Transformers (ViT), CNNs, LLM & Speech Integrations
Data / Cloud	PostgreSQL, MongoDB, Firebase Suite
Tools	Git, GitHub, Tailwind CSS

EXPERIENCE

Full-Stack & Mobile Developer — ANIMMZA

Next.js, React, React Native, Node.js, TypeScript, PostgreSQL, REST APIs *Jan 2025 – Feb 2026 | Hybrid, Lahore, Pakistan*

- Developed full-stack web applications and cross-platform mobile features using Next.js, React, React Native, and TypeScript.
- Engineered scalable backend RESTful APIs, designed PostgreSQL schemas, and implemented secure JWT authentication and RBAC workflows.
- Collaborated across agile sprints to resolve production bugs, optimize state management, and improve application reliability.

Web Development Intern — ARCH TECHNOLOGIES

React, Next.js, Node.js, JavaScript, REST APIs *Aug 2025 – Oct 2025 | Remote, Pakistan*

- Developed frontend UI components and REST endpoints for live web projects using React, Next.js, and Node.js.

SELECTED PROJECTS

GardenMate Ecosystem — FINAL YEAR PROJECT

React Native, Expo, PyTorch (ViT-B/16), Viro React AR, FastAPI, Firebase, Stripe *2025 – 2026*

- Built an AI-powered, AR-integrated gardening assistant and plant marketplace with 12+ specialized modules across four role-based portals (Customer, Vendor, Rider, Admin).
- Fine-tuned a Vision Transformer (ViT-B/16) model for 69 plant species and combined it with the PlantNet API in a 60/40 weighted ensemble, achieving ~350ms inference latency.
- Implemented Viro React AR plane detection for a "Place Before You Buy" 3D plant-placement experience and a camera Snap-to-Autolist product listing workflow.
- Designed the marketplace commission engine with Stripe payment settlement and automated delivery fee calculations across Customer, Vendor, Rider, and Admin portals.

Vouch Keychain — PERSONAL SAAS PRODUCT

Next.js 16, React 19, TypeScript, Firebase Suite, Twilio, Resend, Three.js *2026*

- Built and deployed a QR-based lost-item recovery SaaS product with privacy-preserving masked-contact communication, currently used by around 10 customers.
- Implemented real-time location-recovery telemetry, multi-channel alert dispatch via Twilio SMS/voice and Resend email, and an interactive Three.js 3D product visualizer.

Evershine Academy ERP & LMS — CLIENT PROJECT

Next.js 16, React 19, TypeScript, PostgreSQL *2026*

- Developed ERP/LMS features alongside the team for a deployed multi-campus education system used by 100+ users.
- Built and integrated 7 role-based access portals, automated PDF fee-challan generation, staff payroll workflows, and on-premise biometric fingerprint attendance synchronization.

NutriLens AI

React Native, Expo SDK 54, Zustand 5, Firebase Realtime Database, OpenRouter AI *2025*

- Built a cross-platform mobile nutrition application combining food photography and grocery barcode inputs for automated macronutrient tracking.
- Implemented clinical Mifflin-St Jeor metabolic expenditure calculations (BMR/TDEE) and an interactive AI dietitian coach powered by OpenRouter AI.

FinGo Marketplace

React.js, Node.js, Express.js, MongoDB, JWT *2024*

- Built a multi-vendor commerce platform connecting retail and grocery stores with customers across Lahore, Karachi, and Islamabad.
- Implemented 3-tier role-based access control, secure JWT authentication with refresh-token rotation, and real-time inventory tracking.

Voice AI Copilot

Groq LLaMA 3.3 70B, Groq Whisper Large v3, Web Audio API *2025*

- Built a browser-based voice assistant converting spoken commands into structured task and reminder actions through LLM tool calling.
- Integrated speech recognition and LLM processing using Groq Whisper Large v3 and LLaMA 3.3 70B with the native browser Web Audio API.

ACHIEVEMENTS

Digital Pakistan Speed Programming Competition (DPSPC) **TOP 50 NATIONWIDE**

National Speed Programming & Algorithmic Problem Solving Contest (~450 Teams) *2025*

- Ranked among the top 50 teams nationwide out of ~450 competing teams in speed programming and algorithmic problem solving under strict time constraints.

InnovateX 2.0 Coding Competition **GIFT UNIVERSITY**

Senior-Level Algorithmic Competitive Programming Hackathon *2025*

- Competed in senior-level algorithmic coding challenges evaluating data structures, optimal time/space complexity, and efficient algorithm design.

EDUCATION

Bachelor of Science in Computer Science (BS CS) **GIFT UNIVERSITY, GUJRANWALA**

Coursework: Software Engineering, Data Structures & Algorithms, Database Systems, Artificial Intelligence, Machine Learning, Computer Vision, Distributed Systems *2022 – 2026*